

Midterm project for DATA 37000

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The *Health Insurance Cost Forecasting* dataset was selected for this study because it comprehensively captures real-world factors influencing individual healthcare costs and insurance risks, offering both practical relevance and research value.

This project aims to build and compare multiple classical models, utilizing structured data to forecast key outcomes such as annual medical expenses or high-risk health statuses. The objective is to demonstrate a complete workflow encompassing data preprocessing, feature encoding, model training, and evaluation, while providing actionable insights for practical decision-making in healthcare and insurance analytics.

Dataset and Documentation Description: The dataset used in this project is titled “Health Insurance Cost Prediction,” obtained from the Kaggle platform (source: <https://www.kaggle.com/datasets/mohankrishnathalla/medical-insurance-cost-prediction>).

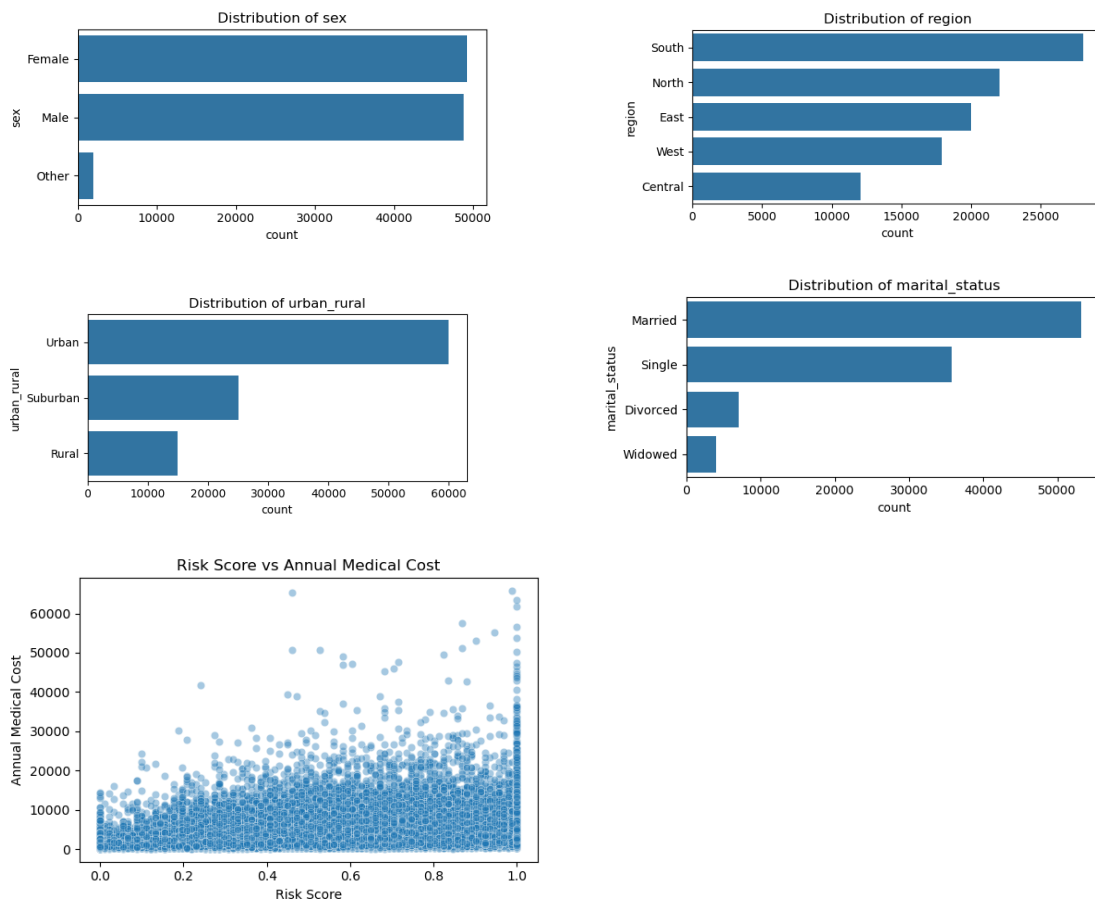
To ensure reproducibility and reference value, the dataset (`medical_insurance.csv`) will also be uploaded to GitHub. This dataset contains 100,000 records and 54 features, covering demographic attributes (such as age, gender, region, education level, and income), lifestyle factors (such as smoking and drinking frequency), medical information (including BMI, blood pressure, chronic conditions, and medical history), and insurance-related variables (such as plan type, premium, deductible, and claims history). Each record represents an anonymous policyholder and contains no personally identifiable information (PII).

For machine learning algorithms: Regression task: Predict annual medical cost (`annual_medical_cost`) as a continuous variable. Classification task: Predict high-risk membership status (`is_high_risk`) as a binary label. This dataset provides a reliable, ethically compliant foundation for exploring data preprocessing, feature engineering, and supervised learning methods introduced in the course.

Exploratory Data Analysis (EDA)

The demographic composition within this insurance dataset and its relationship to risk and cost. By visualizing several key categorical features, we explore their distribution and potential impact on healthcare costs. This dataset exhibits balance, diversity, and information richness.

As shown in the distribution chart, the dataset's content represents mainstream demographics: male and female policyholders are roughly equal in number, with most participants being married, followed by single individuals. This suggests family-related variables may play a significant role in predicting healthcare expenditures.



Regarding residence, the vast majority of policyholders live in **urban areas**, with suburban and rural areas accounting for a smaller proportion. This indicates that urban population data is more representative. **Regional** distribution covers the five major regions (North, South, East, West, Central).

The scatterplot of **risk scores versus annual healthcare costs** reveals a clear positive correlation: higher risk scores correlate with greater healthcare expenditures. As risk scores increase, the cost distribution becomes more dispersed, indicating that high-risk groups not only have higher average expenditures but also experience greater variability in healthcare costs. This observation supports incorporating `risk_score` and health-related variables (`chronic_count`, disease indicators) as strong predictors in subsequent modeling. Overall, exploratory data analysis confirms this dataset's balanced composition, diversity, and information richness, establishing a robust foundation for constructing predictive models—models that link demographic, health, and behavioral factors to insurance risk and cost outcomes.

Model Evaluation:

Three regression models were trained and evaluated on the medical insurance cost prediction task.

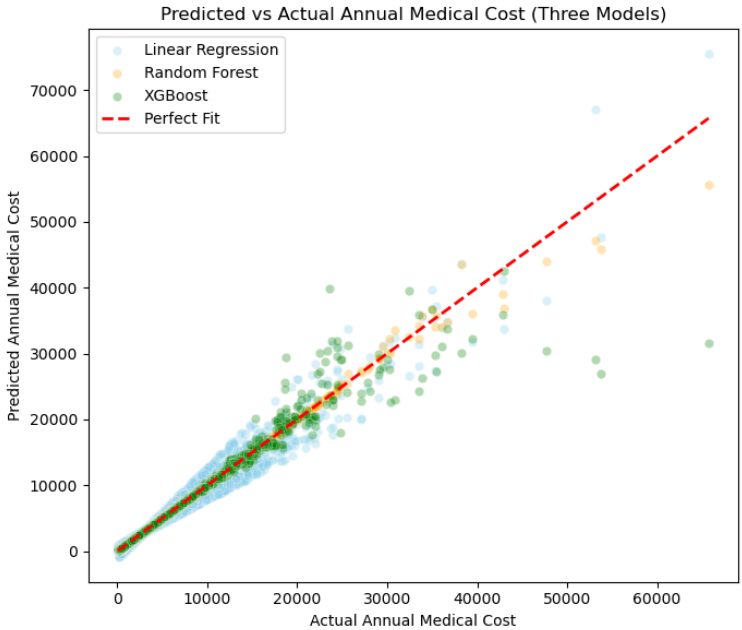
The following metrics were used:

- **R² (Coefficient of Determination)** — measures overall fit
- **RMSE (Root Mean Squared Error)** — measures average prediction error

magnitude

Model	R ²	RMSE
Linear Regression	0.9664	574.83
Random Forest Regressor	0.9982	132.38
XGBoost Regressor	0.9759	486.68

A visualization of predicted versus actual annual medical cost for all three models is shown below.



Three groups of classification models were developed to predict both binary and multi-class outcomes related to healthcare risk and cost levels.

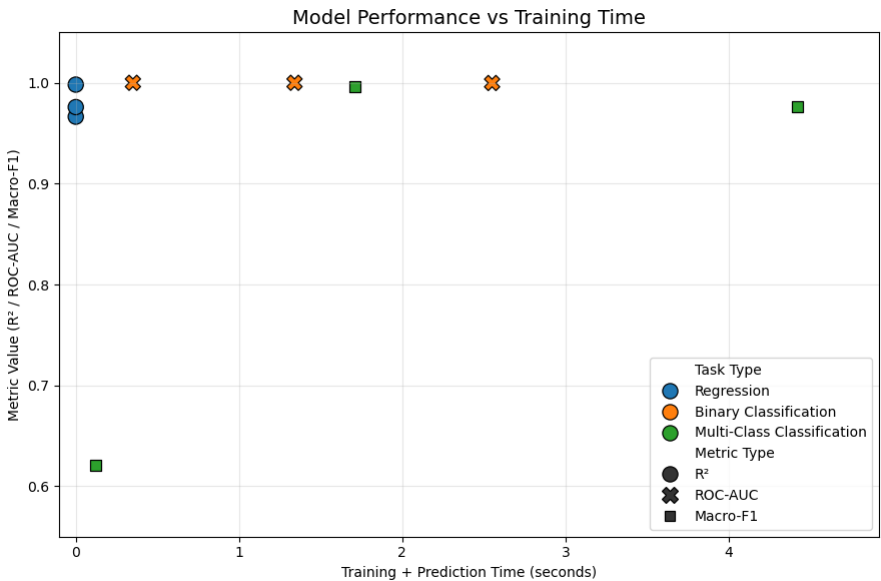
For the **binary task**, Logistic Regression, Linear SVM, and Random Forest were implemented, achieving near-perfect performance across all evaluation metrics. Accuracy, ROC-AUC, and F1 scores were consistently above 0.99, with training times ranging from 0.35 to 2.55 seconds.

For the **multi-class classification**, the models included Multinomial Logistic Regression, K-Nearest Neighbors, and Random Forest.

The Multinomial Logistic and Random Forest models maintained high macro-F1 values above 0.97, while KNN showed notably lower precision and recall due to class overlap. All models completed training and prediction within seconds, with clear differences in runtime efficiency and performance consistency across task types.

Task Type	Model	Main Metric	Value	Time (s)
Binary Classification	Logistic Regression	ROC-AUC	1.0000	1.34
	Linear SVM	ROC-AUC	1.0000	0.35
	Random Forest Classifier	ROC-AUC	0.9998	2.55

Task Type	Model	Main Metric	Value	Time (s)
Multi-Class Classification	Multinomial Logistic	Macro-F1	0.9964	1.71
	KNN (k=9)	Macro-F1	0.6208	0.12
	Random Forest (Multi-Class)	Macro-F1	0.9765	4.42



Discussion

This project compared multiple classical machine learning models across three predictive tasks—regression, binary classification, and multi-class classification—evaluating computational efficiency, interpretability, and probability calibration capabilities.

While most models demonstrated outstanding numerical performance, in-depth analysis revealed significant differences in learning behavior, reliability, and practical applicability.

1. Regression Models

The regression task aimed to predict annual healthcare costs using a blend of demographic, behavioral, and clinical features. The Random Forest Regressor achieved the highest R^2 value (0.9982) and lowest root mean square error, demonstrating its superiority in modeling nonlinear relationships and complex feature interactions. The XGBoost regressor performed slightly below Random Forest but maintained high accuracy and generalization capability, owing to the gradient boosting algorithm's ability to progressively reduce residual error. In contrast, the linear regression model exhibited noticeable bias in high-cost intervals, systematically underestimating extreme expenses due to the limitations of its linear assumption.

These results indicate that ensemble tree-based methods capture nonlinear dynamics

and heteroscedasticity more effectively than linear models. However, for healthcare cost calculations, the computational complexity and limited interpretability of black-box tree models may pose drawbacks, particularly in regulated domains like healthcare and insurance analysis.

2. Binary Classification Models

In binary risk classification tasks, logistic regression, linear support vector machines, and random forests all achieve near-perfect accuracy with ROC-AUC values approaching 1.0. This indicates strong separability between high-risk and low-risk groups given a fixed feature set. However, calibration analysis reveals critical differences in probability reliability.

Logistic Regression models produce steep overconfidence probability curves, assigning extreme probabilities even to borderline cases. While this overconfidence does not affect raw accuracy, it may distort downstream risk assessments. Random Forests and Linear Support Vector Machines (calibrated using Platt's Calibration) both exhibit well-behaved, near-linear calibration curves closely aligned with the ideal diagonal.

From a computational perspective, the linear support vector machine achieved the fastest runtime (≈ 0.35 seconds) while maintaining full accuracy, indicating that a linear boundary suffices in this feature space. Under this characteristic, SVM is the most suitable model.

3. Multi-Class Models

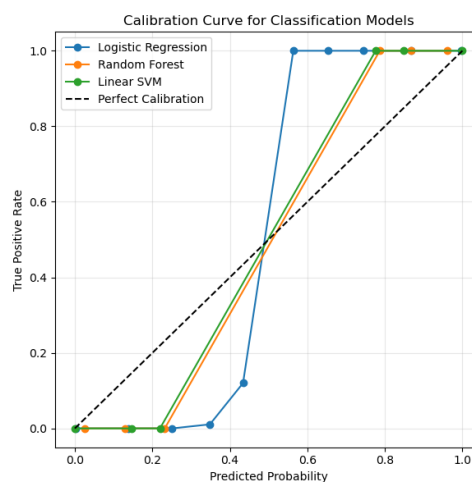
Multi-class cost-graded prediction categorizes samples into low, medium, and high expenditure tiers.

Polynomial logistic regression and random forest classifiers both maintained robust macro F1 scores (0.996 and 0.977, respectively), indicating balanced precision and recall across all cost tiers.

In contrast, the k-nearest neighbors (kNN) algorithm performed significantly worse (macro F1 ≈ 0.62), primarily due to distance-based methods degrading in high-dimensional feature spaces and being sensitive to overlapping category boundaries.

Overall, the multi-class results confirm that both linear and tree-based models effectively capture hierarchical risk patterns, while non-parametric models (e.g., kNN) struggle with structured heterogeneous data.

4. Cross-Task Insights



Several universal patterns emerged across all tasks: Performance vs. Efficiency Trade-off: Ensemble methods (Random Forest, XGBoost) deliver superior predictive capability but require greater computational resources. Linear models (Logistic Regression, Linear Support Vector Machine, Multivariate Logistic Regression) achieve comparable performance at minimal cost, making them suitable for large-scale or real-time inference scenarios. Tree models perform worse than linear models in terms of interpretability and process optimization. Calibration and Probability Reliability: Calibration curve results indicate that well-calibrated models (Random Forest and calibrated SVM) produce probabilities highly aligned with actual event frequencies. This property is critical for decision thresholds in scenarios like health insurance, where predicted probabilities directly translate into financial or clinical actions.

5. Overall Reflection

Comparative analysis demonstrates that no single algorithm comprehensively dominates.

Random Forest excels in accuracy and calibration stability, Linear Support Vector Machines demonstrate significant advantages in efficiency and scalability, while Logistic Regression uniquely stands out for interpretability.

This study reinforces a core principle of applied machine learning: model evaluation must extend beyond accuracy to encompass calibration, computational cost, interpretability, and deployment context. In domains involving real-world decision consequences, the reliability of predicted probabilities is equally critical to the predictions themselves.

Conclusions and Future Work

This project systematically evaluated the performance of various classical machine learning algorithms on regression, binary classification, and multi-class classification tasks using a large-scale health insurance dataset.

Although most models achieved extremely high prediction scores, significant differences existed in their internal behavior, calibration quality, and computational trade-offs. Ensemble tree models such as Random Forest and XGBoost demonstrated optimal predictive accuracy and probability calibration, effectively capturing nonlinear dependencies among demographic, clinical, and financial variables.

Linear models—including logistic regression, linear support vector machines, and multinomial logistic regression—offered faster training speeds and clearer interpretability, making them practical choices for large-scale screening or real-time applications. The findings confirm that performance metrics alone cannot comprehensively evaluate model reliability—when predictions influence financial or clinical decisions, calibration is equally important to interpretability.

Methodologically, this study underscores the importance of holistic evaluation that simultaneously considers accuracy, calibration, efficiency, and fairness. The consistency of calibrated models with the perfect probability baseline demonstrates that even within classical frameworks, model transparency and probabilistic reliability can coexist.

In summary, this study underscores that classical machine learning's strengths lie not only in its simplicity and stability but also in its adaptability to real-world decision systems when combined with robust calibration, interpretability, and governance mechanisms.

Systematic validation, fairness audits, and deployment monitoring will ensure predictive analytics in healthcare and insurance remain accurate and trustworthy.